2022.12.8 习题课

1. 当 $x \neq x'$ 时,

$$\frac{d^2 G(x,x')}{dx^2} + w_0^2 G(x,x') = 0$$

$$\Rightarrow G(x,x') = \begin{cases} A(x') \sin(w_0 x) + B(x') \cos(w_0 x) \\ C(x') \sin(w_0 x) + D(x') \cos w_0 x \end{cases}$$

代边值得, $B(x') = 0$, $C(x') \sin(w_0 l) + D(x') \cos(w_0 l) = 0$

性质, $$\begin{cases} A(x') \sin(w_0 x') = C(x') \sin(w_0 x') + D(x') \cos(w_0 x') \\ w_0 A(x') \cos(w_0 x') - w_0 C(x') \cos(w_0 x') + w_0 D(x') \sin(w_0 x') = 1 \end{cases}$$

再关于 x' 的函数,

$$G(x,x') = \begin{cases} \frac{\sin(w_0(l-x'))}{w_0 \sin(w_0 l)} \sin w_0 x, & x \leq x' \\ \frac{\sin w_0 x'}{w_0 \sin(w_0 l)} \sin(w_0(l-x)), & x > x' \end{cases}$$

定解问题: $$a \frac{\sin(\ )}{\sin w_0 l} + b \frac{\sin w_0 x}{\sin(w_0 l)}$$

交叉乘减,积分,

$$\int (y G'' - G y'') dx = \int (y\delta - f(x)) dx$$

$$\Rightarrow (yG' - Gy')\Big|_0^l = y(x') - \int_0^l f(x) G(x,x') dx$$

代边值得:

$$y(x) = \frac{b \sin w_0 x + a \sin(w_0(l-x))}{\sin w_0 l} + \int_0^l f(x') G(x,x') dx'$$

3、按本征函数展开，

$$G(x,y;x',y') = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} A_{mn} \sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

代入泊松方，

$$\int_0^a\int_0^b \nabla^2 \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} A_{mn}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b} = \iint -\delta(x-x')\delta(y-y')\,dx\,dy$$

正交性，

$$A_{mn} = \frac{2}{ab}\,\frac{-1}{(\frac{m\pi}{a})^2+(\frac{n\pi}{b})^2}\sin\left(\frac{m\pi x'}{a}\right)\sin\left(\frac{n\pi y'}{b}\right)$$

$$\Rightarrow \quad G(x,y,x',y') = \frac{2}{ab}\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{-1}{(\frac{m\pi}{a})^2+(\frac{n\pi}{b})^2}\sin\left(\frac{m\pi x'}{a}\right)\sin\left(\frac{n\pi y'}{b}\right)\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)$$

交叉乘减，

$$\nabla^2 G u - \nabla^2 u G = -u\delta\delta - fG$$

$$\Rightarrow \quad \oint(\nabla G u - \nabla u G)\,d\vec{s} = -u(x',y') - \int_0^a\int_0^b f(x,y)\,G(x,y,x',y')\,dx\,dy$$

代入，
$$u(x,y) = \frac{2}{ab}\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{1}{(\frac{m\pi}{a})^2+(\frac{n\pi}{b})^2}\left\{\left[\frac{m\pi}{a}\int_0^b \sin\frac{n\pi y'}{b}\left(\alpha(y')-(-1)^m\beta(y')\right)dy' - \frac{n\pi}{b}\int_0^a \sin\frac{m\pi x'}{a}\left(\mu(x')-(-1)^n\nu(x')\right)dx'\right]\right.$$
$$\left.+\int_0^a\int_0^b \sin\left(\frac{m\pi x'}{a}\right)\sin\left(\frac{n\pi y'}{b}\right)\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)f(x',y')\,dx'dy'\right\}$$